

CNC-X100 Field Service & Overhaul Manual

Product: CNC Machine | Model: CNC-X100 | Document Type: SERVICE_MANUAL

1. Service Overview & Certified Technician Qualification

This Service Manual provides authorized field service procedure standards for ProductAssist technicians servicing the CNC-X100 vertical machining center.

- Service Clearance: Maintain 1.0-meter open clearance around rear electrical cabinet and side pump enclosures during field servicing.
- Zero Energy State: Always execute Lockout/Tagout (LOTO) and verify 400V 3-phase bus voltage reads zero with calibrated meter prior to touching internal wiring.

2. Component Access & Service Inspection Points

- Coolant Tank & Intake Screen: Accessible at lower rear cabinet. Screen basket mesh size 0.5 mm requires flushing when flow drops below 8.0 L/min.
- Spindle Drive Motor & Bearing Chiller: Located at top headstock enclosure. Chiller unit requires annual refrigerant pressure check.
- Z-Axis Ball Screw & Linear Guides: Remove upper accordion way cover. Lubricate with ISO VG 68 way oil every 500 hours.
- Electrical Cabinet Breakers & Relays: Located on rear door panel. Check thermal overload relay Q4 (coolant pump) setting: 8.5 Amps.

3. Major Service Procedures & Field Overhauls

- Coolant Pump Replacement: Disconnect 400V 3-phase pump leads, unbolt flange bolts, replace pump assembly (Part # X100-PMP-001).
- Spindle BT40 Drawbar Retention Spring Replacement: Use hydraulic spring compressor tool to replace Belleville washer stack (retention force 9.5 kN).
- Axis Ball Screw Backlash Adjustment: Adjust pre-load double nut until backlash reads <0.003 mm on dial indicator.

4. Field Service Escalation Conditions

Escalate to Senior Systems Engineer immediately if:

- Spindle bearing thermal runout exceeds 0.015 mm or vibration exceeds 4.5 mm/s.
- Main casting structure exhibits cracking or foundation anchor bolt displacement.
- Short circuit faults damage internal 24V DC I/O bus controller cards.